ZHIQI HUANG

FUKUOKA, JAPAN

EMAIL HOMEPAGE SCHOLAR GITHUB

RESEARCH PROFILE

- M.Phil. student at Waseda University working on geometry-aware generative modeling across 3D assets, PBR materials, 3D Gaussian Splatting, and 3D-aware video/world models.
- Seeking RA and 2027 Ph.D. opportunities in graphics, vision, generative AI, embodied simulation, controllable digital humans, and efficient multimodal 3D generation.

RESEARCH INTERESTS

- Geometry-aware generative modeling with depth, normal, material, and multi-view priors for 3D, video, and world models.
- 3D-aware video/world models for embodied interaction, robot manipulation, AR/VR, and interactive simulation.
- Simulation-ready assets and digital humans: relightable PBR materials, 3DGS avatars, controllable objects, and consumer-GPU generation.

EDUCATION

Waseda University — Apr. 2025–Mar. 2027 (expected)

M.Phil. in Information Architecture — Fukuoka, Japan

- English-taught program
- GPA: 3.8 / 4.0
- Research focus: geometry-aware generative modeling, 3D generation, and world models for embodied interaction

Sun Yat-sen University — 2017–2021

B.E. in Software Engineering — Guangzhou, China

- GPA: 3.7 / 4.0

PUBLICATIONS

SIE3D: Single-Image Expressive 3D Avatar Generation via Semantic Embedding and Perceptual Expression Loss — 2026

IEEE ICASSP 2026 (Accepted); First Author, Corresponding Author

- Authors: Zhiqi Huang, Dulongkai Cui, Jinglu Hu
- Fuses image identity embeddings with text semantic embeddings and a perceptual expression loss for expression-faithful, language-controllable 3DGS avatar generation.
- Links: Project Page, Paper, Code

MANUSCRIPTS

First-author manuscript on geometry-aware PBR material generation from long text — 2026

First Author; ACM MM 2026 (Under Review)

- Generates relightable PBR maps from long text by grounding material semantics in local geometry with mesh-normal priors.
- Evaluates semantic alignment, reconstruction quality, multi-view consistency, and consumer-grade GPU inference for simulation-ready 3D assets.

First-author manuscript on deform-then-edit lung-nodule CT forecasting 2026

First Author; Target: AAAI 2027

- Studies deform-then-edit 3D forecasting for longitudinal CT, using bounded local deformation fields, mask/shell shape priors, and gated residual edits to predict future ROI appearance while preserving background anatomy.

First-author manuscript on text-conditioned PBR material generation 2026

First Author; Target: AAAI 2027

- Maps material prompts into MaterialMVP/Hunyuan3D-Paint reference-condition features, replacing image-derived DINOv2 features with text-generated material conditions for geometry-aware PBR texture diffusion.

INDUSTRY EXPERIENCE

4399 Games 2023–2024

Senior Graphics Engineer Guangzhou, China

- Led a 3–5 person rendering team and owned the rendering roadmap for *Era of Conquest* on mobile and PC.
- Drove graphics development for new mobile, PC, and web projects, connecting production rendering workflows with practical generative systems for games and simulation.

4399 Games 2021–2023

Graphics Engineer Guangzhou, China

- Built and optimized the real-time rendering pipeline for *Era of Conquest*, focusing on cross-platform shader optimization, PBR, and performance profiling.

TECHNICAL SKILLS

- Programming & graphics: C++, Python, C#, GLSL/HLSL; Vulkan, OpenGL, Unity, real-time rendering, physically based rendering (PBR).
- ML & 3D research: PyTorch, 3D Gaussian Splatting, diffusion models, CLIP/LongCLIP, mesh processing, PBR material generation, parameter-efficient fine-tuning, multi-view evaluation.

LANGUAGES & HONORS

- Chinese: Native (Mandarin and Cantonese)
- English: Professional working proficiency (TOEFL iBT: 90)
- Outstanding Student Scholarship (Third Prize), Sun Yat-sen University, 2018–2019